Part D

1.Which Model Is Best Between Model 1 and Model 2

The model that performs best should be the one with the lowest Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), together with the highest R^2 score. These measures help show how close the predictions are to the actual Length of Stay values.

If the Part C results show that XGBoost Regression performed better than Random Forest Regression, then XGBoost would be the preferred option. One of its strengths is that it can identify complex, non-linear relationships between variables. It uses a gradient boosting approach to improve its predictions over a series of training rounds (Jain et al., 2024).

XGBoost works by learning from errors made in earlier rounds and using that information to improve later predictions. This can help it identify more complicated patterns in healthcare data that may affect how long a patient stay's in hospital. Jain et al. (2024) similarly discuss the value of boosted tree models for predicting healthcare outcomes, including Length of Stay.

Random Forest Regression also showed that it can make useful predictions. However, if XGBoost achieved the lower prediction errors and higher R^2 score in Part C, it would be the stronger candidate for STADIOcare. More accurate Length of Stay estimates could help managers make better-informed decisions about staffing, bed availability and the movement of patients through the hospital.

For these reasons, XGBoost Regression is recommended as the main model for further development, provided that its evaluation results in Part C confirm the stronger performance.

2. How Can the Selected Model Be Improved

The selected model provides a useful starting point, but there are several ways it could be improved before it is considered for use in a hospital setting.

One improvement would be to include a wider range of healthcare information. Length of Stay can be affected by a patient's condition, treatment and circumstances, as well as by operational factors. Diagnosis categories, procedures, laboratory results, comorbidities, patient acuity and previous admission patterns could therefore give the model more information to work with. Jain et al. (2024) also highlight the value of using a broader range of healthcare variables when predicting Length of Stay.

Another area to consider is hyperparameter tuning. The current model uses predefined settings, which may not be the best combination for this dataset. Grid Search, Random Search or Bayesian Optimisation could be used to test different settings and identify combinations that improve the model's performance.

The feature engineering could also be developed further. For example, occupancy rates, appointment demand, seasonal admission patterns, staffing levels and patient-flow information may help explain changes in hospital demand. This is in line with Cahan et al. (2024), whose research highlights the value of using patient-level information together with operational data when forecasting hospital occupancy.

The model should also be reviewed and retrained regularly. Patient profiles, disease patterns and treatment practices can change over time, which may affect how well the model performs. Using more recent data for retraining would help keep its predictions relevant.

It may also be useful to test ensemble methods, which combine predictions from more than one algorithm. This could make the results more reliable and help reduce variation in predictions, although the benefit would need to be confirmed through further testing.

3. How Will the Selected Model Need to Be Adapted to Work with the Requested Data from SS1

MIMIC-IV was useful for developing and testing the models, but it is not a substitute for STADIOcare's own data. Before the selected model could be used by the group, it would need to be adapted to reflect the organisation's hospitals, patients and operating conditions.

The first step would be to identify how STADIOcare's data relates to the variables used by the model. The SS1 case study identified several relevant sources, including admissions, occupancy information, workforce records, labour costs, patient appointments and other operational measures. These data would need to be collected, cleaned and brought together in a consistent format before they could be used for modelling.

Data quality would need particular attention. As noted in the SS1 scenario, STADIOcare faces challenges with the quality, completeness and historical coverage of its data. Missing values, inconsistent entries and duplicate records should therefore be identified and handled during preprocessing.

The model would then need to be retrained using STADIOcare's own historical data. A model developed on one dataset may not reflect the patterns found in another healthcare organisation.

The model could also be strengthened by including features that reflect the operational pressures identified in the STADIOcare case study. These could include:

- Hospital occupancy rates
- Bed utilisation percentages
- Agency staffing expenditure
- Overtime costs
- Seasonal admission trends
- Appointment demand

A further requirement would be to establish a reliable data pipeline. This would allow new information to be added as it becomes available and support regular model updates. With appropriate checks in place, this could help ensure that predictions remain relevant to day-to-day capacity planning.

4. Do the Results Align with the Related Literature from Part A

Overall, the findings from this project are broadly in line with the research reviewed in Part A.

Jain et al. (2024) examined the use of models such as Random Forest, XGBoost, LightGBM and CatBoost to predict hospital Length of Stay. Their study reported that tree-based machine learning methods can perform better than traditional regression approaches in healthcare prediction tasks. The results from this project, where both Random Forest and XGBoost showed predictive ability, are consistent with that work.

Alibrahim et al. (2023) highlighted how predictive analytics can support hospital bed-capacity planning and resource management. Their research points to the value of forecasting for decisions about healthcare demand, occupancy and daily operations. The work in this project is relevant to those findings because estimating Length of Stay can help hospitals anticipate how long beds may be occupied and plan staffing and other resources accordingly.

Cahan et al. (2024) also found value in combining patient-level information with operational data when forecasting hospital and ward occupancy. The feature engineering in this project follows a similar approach by using admission information alongside additional operational indicators.

Taken together, the findings are consistent with the literature reviewed in Part A. Both models showed potential for predicting Length of Stay, while XGBoost is the preferred option if its performance measures

in Part C are stronger than those of Random Forest. This supports the wider view in the literature that machine learning can contribute to better operational planning and resource management in healthcare. However, STADIOcare would still need to test the model using its own data before relying on its predictions.

Recommendations

The following actions are recommended for STADIOcare based on the work completed in this project:

1.Use XGBoost Regression as the main model for predicting Length of Stay, provided that the Part C evaluation confirms it achieved the strongest results.

2.Retrain the model using STADIOcare's historical admissions, occupancy, staffing and operational data.

3.Include additional indicators such as hospital occupancy, labour costs, bed utilisation and appointment demand.

4.Monitor the model's performance regularly so that any decline in prediction accuracy can be identified and addressed.

5.Develop reliable data pipelines to support regular updates and retraining of the model.

6.Use the model's predictions to inform workforce planning, bed allocation and wider resource-management decisions.

If these recommendations are implemented carefully, STADIOcare could improve its planning, make better use of available resources and support decisions with more reliable evidence across its hospital network.

References

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